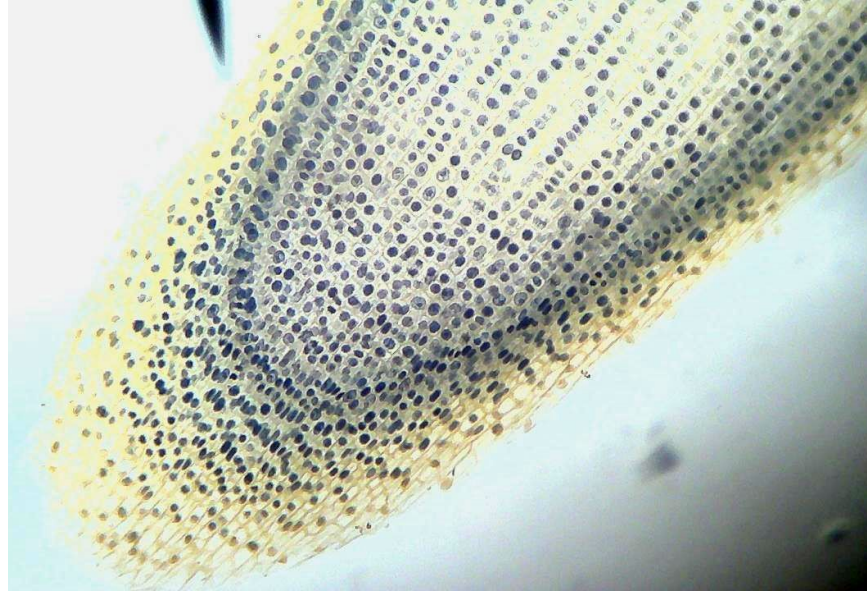




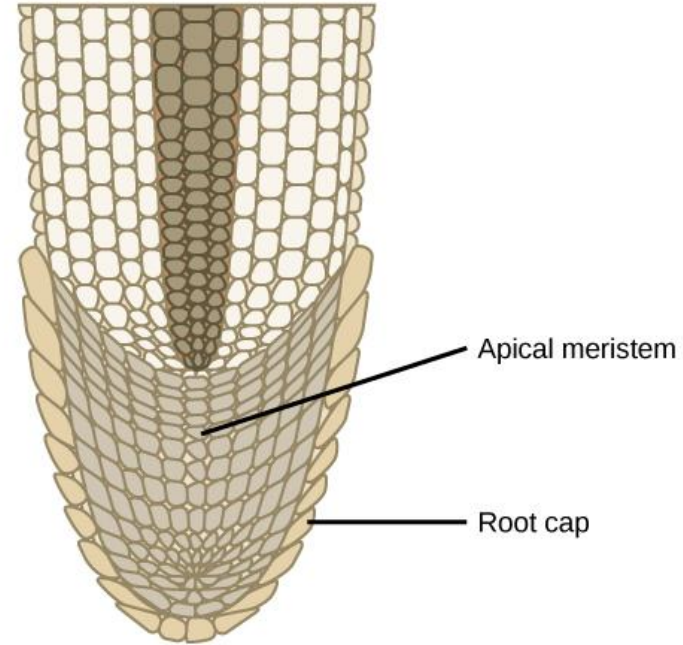
Investigation: Mitosis

(remote edition)

Introduction

One of the basic tenets of biology is that all new cells come from living cells. New cells are formed by the process of cell division which includes both the division of the cell's nucleus, **mitosis**, followed by cytokinesis. Cell divisions are used by organisms to repair tissue, growth, and asexual reproduction. Cells created in this process are identical to the original.

Where does one find cells undergoing mitosis? It depends if you are studying plants or animals. Plants and animals differ in this respect. In most plants, the process of forming new cells is restricted to special growth regions called **meristems**. These regions usually occur at the tips of the stems and roots of the plant. Mitosis in animals can be viewed by looking at the growing embryo or at cells of tissue that rapidly divide.



1. Where do you look in a plant cell to find dividing cells?

2. Why do cells divide?

The Cell Cycle

The cell spends a majority of its life in **interphase**, where it performs cell processes and copies its DNA. In interphase, the nucleus will appear as a dark structure near the center of the cell.

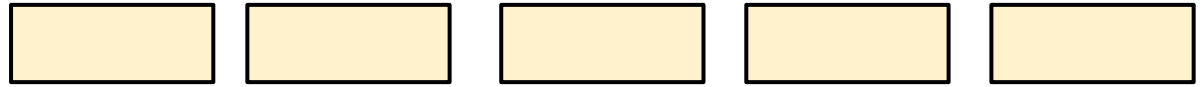
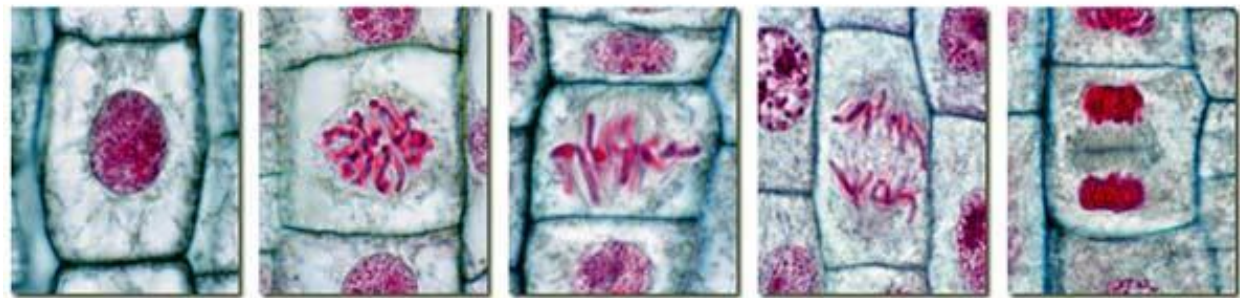
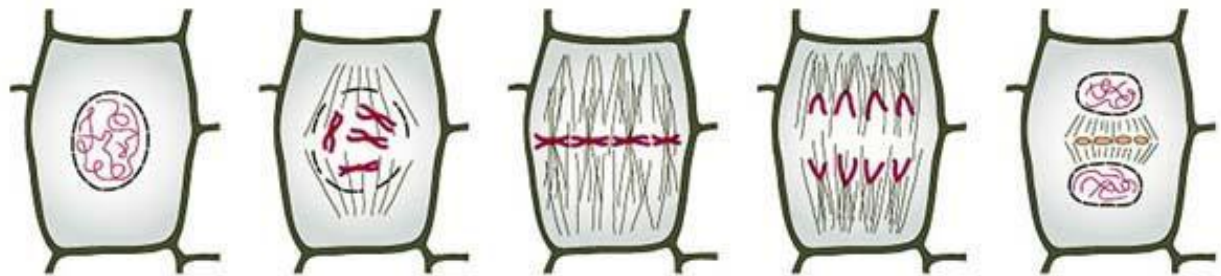
Mitosis begins with **prophase**, where the DNA in the nucleus condenses and chromosomes can be seen. The spindle forms, which moves chromosomes.

During **metaphase**, the chromosomes line up at the equator of the cell.

Chromatids (identical copies of chromosomes) separate during **anaphase**.

The last phase is **telophase**, where the nuclear membrane reforms around two new daughter cells and cytokinesis begins.

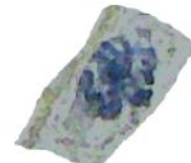
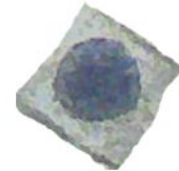
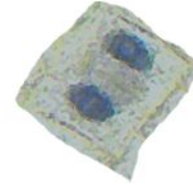
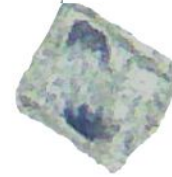
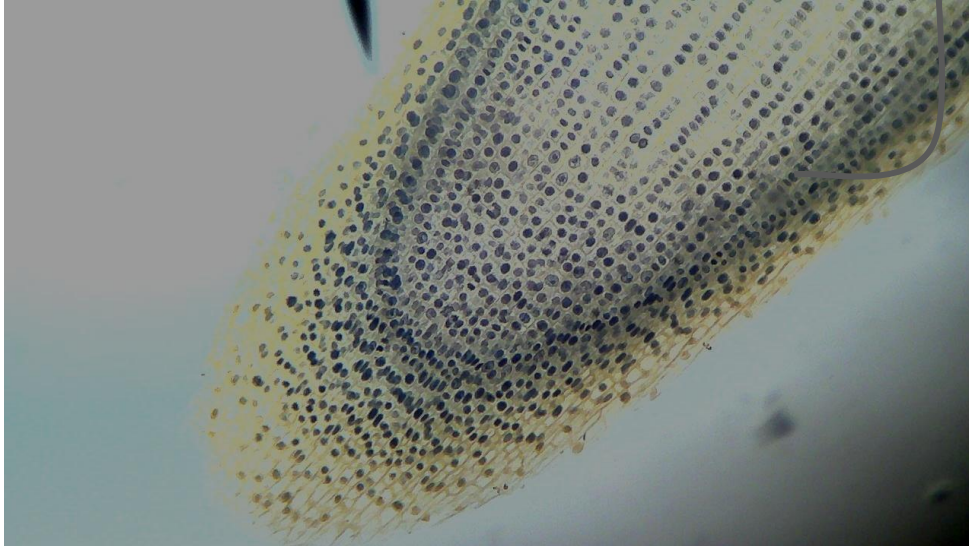
3. Label the phases of the cell cycle:



The images show (in no particular order) photos from a microscope slide of an onion root.

4. Label the phases.

5. Which phase do you think is the hardest to identify? Why?



How much time is spent in each phase of the cell cycle?

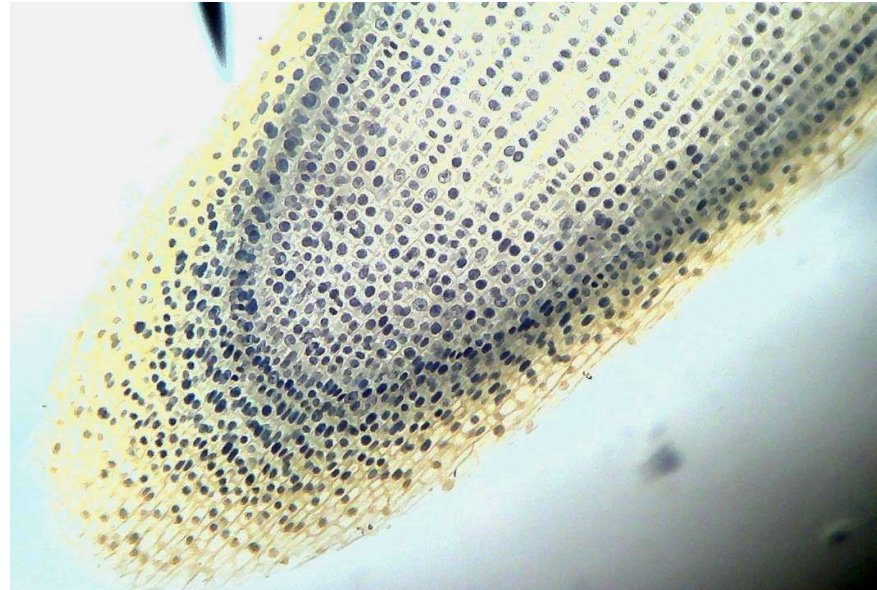
You will have slides of onion root tips where cells are undergoing mitosis. The slides are prepared by “fixing” the cells and staining them. They are no longer dividing, but trapped in time.

To determine the amount of time the cell spends in each phase, you will count the number of cells that are in observed in each phase. The total time for the cell cycle in an onion cell is 720 minutes (12 hours). To calculate the amount, this formula can be used:

$$\frac{\text{Number of cells in a specific phase}}{\text{Total number of cells counted}} \times 720 \text{ minutes} = \text{time for specific phase}$$

6. This is an onion root tip at low power (100x). The cells are too small to distinguish individual phases. What phase do most of the cells appear to be in?

7. Make a prediction about what phases will be longest and what phases will be shortest:



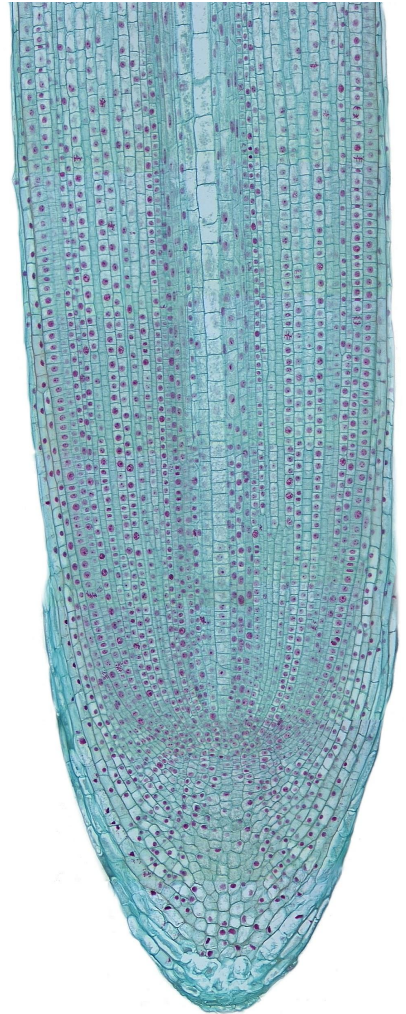
8. Access the cell images at this [Google Drive Folder](#) (or scroll to the last slides in this document) and complete the table by counting the number of cells in each phase. This may be challenging and some cells you might need to make a decision about.

Stage of Mitosis	# Cells in Slide 1	# Cells in Slide 2	# Cells in Slide 3	# Cells in Slide 4	# Cells in Slide 5	Total # Cells in this phase	Time (use formula)	% of Cell Cycle (Time/720 x 100)
INTERPHASE								
PROPHASE								
METAPHASE								
ANAPHASE								
TELOPHASE								
$\frac{\text{Number of cells in a specific phase}}{\text{Total number of cells counted}} \times 720 \text{ minutes} = \text{time for specific phase}$						Total # of calls counted from all slides	720	100%

9. What is the longest phase of the cell cycle. Suggest a reason for this to be the case.

10. Are the other phases of mitosis equal (or close)? Suggest a reason for any differences in the timing of mitosis phases.

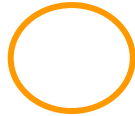
11. You were asked in the beginning to suggest which phase would be the most difficult. Now that you've viewed cells in mitosis, what did you find difficult about completing the table on the previous slide?



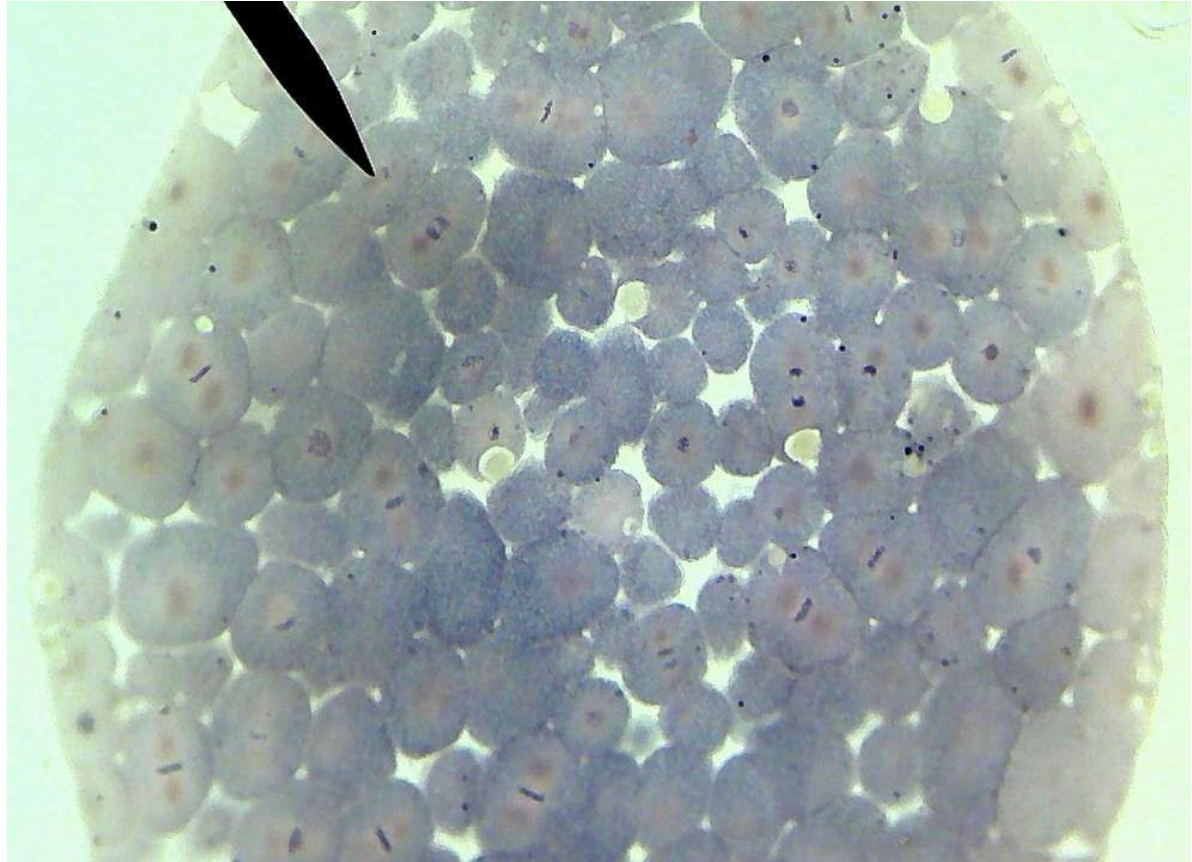
The image shows the blastula of a whitefish. A blastula is an early embryo that is growing rapidly.

12. Examine the blastula slide. How does it differ from meristem slide?

13. What phase of mitosis can you find on the slide. Use the circle tool to identify it on the image?



14. How do the number of cells in interphase compare to that of the onion? Suggest a reason for this difference.



15. Final Synthesis. Describe how a cell that is FIXED can be used to estimate the time spent in the phases of mitosis. Explain the process you used to determine the times for each phase.

